Steps involved in SCP production

Provision of carbon source with added nitrogen, CO_2, ammonia, trace minerals for growth

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Prevention of contamination by using sterilized medium and fermentation equipments

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Selected microorganism is inoculated in a pure form

↓

Adequate aeration and cooling is provided

↓

Microbial biomass is harvested and recovered by flocculation or centrifugation flocculants

↓

Harvested algae are dewatered and dried on open sand beds

↓

Processing biomass and enhancing it for use and storage

Advantages of using microorganisms for SCP production:

1. Microorganisms grow at a very rapid rate under optimal culture conditions.
2. The quality and quantity of protein content in microorganisms is better compared to higher plants and animals.
3. A wide range of raw materials which are otherwise wasted, can be fruitfully used for SCP production
4. The culture conditions and the fermentation processes are very simple.
5. Microorganisms can be easily handled and subjected to genetic manipulations.

Table 6.6: List of microorganisms and substrates used for SCP production

Microorganisms	Substrates
Bacteria	
Pseudomonas sp.	Alkanes
Methylomonas sp.	Methanol
Yeast	
Candida utilis	Sulfite liquor
Lactobacillus bulgaricus	Whey
Fungi	
Aspergillus niger	Molasses
Trichoderma viridae	Straw, starch
Algae	
Spirulina maxima	Carbon di oxide
Scenedesmus acutus	Carbon di oxide
Actinomycetes	
Nocardia	Alkanes
Mushroom	
Agaricus bisporus	Compost, rice straw
Volvariella volvacea	Cotton straw

During the cultivation of SCP, care must be taken to prevent and control the contamination by other micro organisms, which produce mycoxins or cyanotoxins. This is controlled by using the fungus *Scytaliclium acidophilum* which grows at a low PH. It allows the hydrolysation of paper wastes to a sugar medium and also creates aseptic condition at low cost.

6.9 Industrial Production of Citric Acid

Citric acid is obtained from citrus fruits; pineapple etc., and after the development of microbial fermentation, citric acid production becomes very cheap, easy and cost effective. 70% of citric acid produced is used in food and beverage industry. Many microbial strains such as fungi *Aspergillus flavus, Aspergillus niger* and *Trichoderma viridae*, yeast *Hansenulla polymorpha*

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